

Mohamed Khalil

Front-End Developer – Angular, TypeScript
+20 111 999 2330 | mohamed.khalil.ucbl@gmail.com | Egyptian and French citizen (EU work authorization)
linkedin.com/in/mohamed-khalil-ucbl | github.com/Mohamed-Khalil67 | portfolio-swart-omega-11.vercel.app

Summary

Software Developer with 5+ years of engineering experience who moved from real-time embedded systems (70% cost reduction and 10x better precision on an adaptive lighting system) into web development, now building production Angular and TypeScript applications. Driven by curiosity and a passion for software, I integrate AI into my daily work, collaborate well in teams and lead ideas from concept to delivery.

Experience

Full-Stack R&D Developer (Fixed-Term Contract) Oct 2025 – Mar 2026
Dedalus – WebSuite hospital information system *Bordeaux, France*
Tech stack: JavaScript (ES6+), HTML/CSS, XML, Delphi (Object Pascal), Oracle PL/SQL, REST APIs, Scrum

- Enabled real-time editing of clinical PDFs with a vanilla JavaScript/XML module on the in-house SJS framework
- Streamlined clinicians' patient-data entry with a contextual, interactive panel that improved data display and workflow
- Collaborated in a Scrum team to fix full-stack incidents with Delphi functions and Oracle PL/SQL stored procedures

R&D Embedded Engineer Jan 2024 – Jan 2025
Johnson Electric (AML Group) *Le Bourget, France*
Tech stack: Embedded C, LabVIEW, MEMS sensors, DC motor control, PID control

- Achieved a 70% cost reduction and 10x better positioning precision by refining algorithms and calibrating hardware
- Cut headlamp auto-adjustment latency by redesigning interrupt-driven routines to raise MEMS sampling frequency
- Eliminated beam overshoot during sharp braking by developing and tuning a closed-loop PID controller
- Sped up beam-levelling debugging by reverse-engineering MLX81310 driver firmware for LabVIEW integration

Test and Validation Engineer Sep 2022 – Jan 2024
Valeo *Créteil, France*
Tech stack: CANoe, CANalyzer, CAN bus, ADAS ultrasonic sensors

- Improved ADAS ultrasonic sensor accuracy on Renault vehicles by calibrating sensors and parameters in CANoe
- Verified 3 ADAS functions (Auto Park, object detection, emergency stop) via functional tests and CANalyzer bus analysis
- Aligned optimal sensor positioning with the client while tracking specifications on Renault prototypes

R&D Embedded Engineer Sep 2020 – Aug 2022
Valeo *Créteil, France*
Tech stack: Embedded C, C++, PHP, Arduino, RTMaps, Lidar, ArUco, Protocol Buffers

- Enabled autonomous navigation on a robotic platform with Arduino odometry in embedded C
- Delivered camera-based 3D object detection with custom C++ RTMaps modules and real-time Lidar/camera visualization
- Achieved accurate marker recognition via ArUco tracking; built a PHP web interface over Protocol Buffers

Projects

Hoodizz – E-commerce Platform | *Angular 21, NgRx SignalStore, NestJS, Prisma, Stripe* Mar 2026 – Jun 2026

- Shipped a full-stack TypeScript e-commerce platform in an Nx monorepo: an Angular 21 storefront (standalone components, Signals, SSR, incremental hydration, zoneless, SCSS) and a NestJS API on Prisma/PostgreSQL
- Prevented overselling and spoofed payments with 3 REST endpoints, Prisma stock transactions, verified Stripe webhooks and 23 Jest suites

Portfolio – CSS Landing Pages | *Angular 21, SCSS, HTML5, CSS3* May 2026 – Jul 2026

- Hand-coded 6 landing pages (e-commerce, finance app, product, blog, gaming) in HTML5 and CSS3 with Flexbox, Grid, custom properties, clamp() and keyframe animations
- Built the Angular 21 portfolio site that showcases them, using standalone components, Signals and SCSS, deployed on Vercel

Source and live demos: github.com/Mohamed-Khalil67/Hoodizz | hoodizz-retail-markt.vercel.app | github.com/Mohamed-Khalil67/Portfolio

Education

H3 Hitema Créteil, France
M.Sc. Information Systems Design and Development *2022*

- Coursework: cross-platform development (web, mobile, desktop, IoT), multi-level testing and code review, CI/CD pipelines

Université Claude Bernard Lyon 1 Lyon, France
M.Sc. Electronics and Embedded Systems (2020); B.Sc. Electronics, Electrical Energy and Automation (2018) *2018 – 2020*

Technical Skills

Languages: TypeScript, JavaScript (ES6+), HTML5, CSS3, SCSS, SQL (PostgreSQL, Oracle PL/SQL), Python, C/C++
Front-End: Angular, RxJS, NgRx, NgRx SignalStore, Angular Signals, standalone components, SSR, hydration, Tailwind CSS, Bootstrap, responsive design, web accessibility
Back-End and APIs: NestJS, Node.js, REST APIs, Prisma, PostgreSQL, Firebase Auth, OAuth 2.0, Stripe
Testing, Tools and Methods: Jest, Git, GitHub, GitLab, Nx, Docker, Google Cloud Platform (GCP), Vercel, Railway, Figma, Agile/Scrum
AI Skills: CI/CD pipelines, AI integration and workflow automation (n8n), building MCP servers, creating project-specific AI-tool skills
Soft Skills: communication, teamwork, problem solving, client coordination, adaptability, time management
Spoken Languages: Arabic (native), English (fluent, TOEIC B2), French (fluent, DELF B2)